

The face inversion effect in log-polar models

1. The data

Established clear line - Test and training identities never mix

Training categories.

category	what it is	size
faces	CelebA identities	128
faces_vgg	VGGFace2 identities — natural photo variation	480
faces_cfd	CFD identities — studio, neutral background	119
faces_rfwW / faces_rfw0	RFW identities, white / other-race	1200 / 300
objects	object classes	64
houses_zubud	ZuBuD buildings, each its own identity	40 or 137
houses_gen / houses	buildings lumped into one generic house label	635–732

Test stores. Held out of training, so the model has never seen these identities:

store	what it is
faces_cfdWM64	64 white-male CFD faces. Studio-lit, plain background, so identities differ mainly in facial structure. This is where we read face effects.
faces_rfwWM64	64 white-male RFW faces, in the wild. Used to calibrate, not to measure — see §4.
faces_vggH0	60 novel VGGFace2 identities
houses_yin64	64 ZuBuD buildings
objects	the object set

The two tasks. Yin (1969) is a memory task: study 40 items, then 24 old-vs-new 2AFC pairs. Dobs/Kanwisher (2023) is a matching task. One noise parameter **p** per model — bit-flip noise on the 256-unit code — is tuned so upright accuracy sits on the human anchor. Because **p** differs between models, **compare effects, not raw accuracies.**

Human targets.

Yin (1969)	upright accuracy	inversion effect
Faces	96.29	+ 14.41
Houses	90.71	+4.96
Objects	84.79	+0.83

2. The models, before we touched orientation

All ResNet-18, trained from scratch, log-polar crops around 16 fixation points. Six-stage curriculum, 80 epochs. **No inverted images in training** — a deliberate rule, so the inversion test stays clean.

model	face diet	face ids	objects	house ids	generic house class	total classes
r15_vgg	CelebA + VGGFace2 + CFD	727	64	40 ZuBuD	635 HD	832
r16_base	CelebA + VGGFace2	608	64	40 ZuBuD	635 HD + 97 ZuBuD	713
r16_rfw	r16_base + 1500 RFW	2108	64	40 ZuBuD	635 HD + 97 ZuBuD	2213
r17_base	CelebA + VGGFace2	608	64	40 ZuBuD	635 HD	713
r17_houses	CelebA + VGGFace2	608	64	137 ZuBuD	635 HD	810
r16rfwftH	fine-tune of r16_rfw	2108	64	137 ZuBuD	635 HD	2310

Each face and house identity is its own class, and each object class is one class. The generic house class is **one label no matter how many buildings it holds** — 635 or 732 — which is what §3 turns out to be about. Note how lopsided the diet is: even the smallest face arm has 608 identities against 40 houses.

The faces got better. Adding VGGFace2 gave r15 a Kanwisher result within half a point of humans, on identities it had never seen:

	upright	inversion effect
Human	87.50	+10.70
r15_vgg on faces_vggH0	86.10	+11.10

Adding RFW made it better still. On held-out CFD faces, r16_rfw shows an upright-over-inverted advantage of **+16.25** against a human +14.41. The same model without RFW reads +10.75. 1500 in-the-wild identities seem to be what builds a face representation that cares about orientation.

3. The houses fell over, and we fixed them

While the faces improved, the house control collapsed. Both r16 models scored held-out ZuBuD buildings **below the 50% chance floor**:

model	generic house class	house ids	houses_yin64 upright
Human	—	—	90.71
r16_base	HD + 97 ZuBuD	40	33.92
r16_rfw	HD + 97 ZuBuD	40	42.17
r16rfwftH	HD only	137	59.92
r17_base	HD only	40	85.17
r15_vgg	HD only	40	87.75
r17_houses	HD only	137	96.50

The cause. r16 folded 97 leftover ZuBuD buildings into the single generic house label. Training the network to map 97 ZuBuD buildings onto one label destroys the representation the other ZuBuD identities depend on.

The fix. Take ZuBuD out of the generic class (r17_base: 85.17). Add more house identities on top, 40 → 137, and it goes to ceiling (r17_houses: 96.50). A 20-epoch fine-tune recovers most of it without retraining from scratch (r16rfwftH: 59.92).

Two things this leaves behind:

- **More house training makes houses behave more like faces.** r17_houses (137 ids) shows a *bigger* inversion cost than r17_base (40 ids). The cost tracks how much individuation

training a category got, not what kind of category it is. That weakens any face-specificity claim resting on houses.

- **Houses are memorised, not learned.** 120 house images take 10% of every batch, so each crop is seen $\sim 11,909$ times over the run against 60 for `faces_vgg`. The weight is compensating for a dataset far too small.

4. The main finding: the model has good memory, bad orientation sense

Yin’s design has four cells — study orientation, then test orientation. UU, UI, IU, II. Earlier work here only reported UU minus UI. Running all four shows that was the wrong number.

`faces_cfdWM64`, one calibrated p per model, 50 seeds:

model	p	UU	II	UI	IU
Human	—	96.29	81.88	84.13	78.58
<code>r15_vgg</code>	0.09	89.17	85.17	69.50	63.25
<code>r16_base</code>	0.08	85.25	74.50	63.83	63.67
<code>r16_rfw</code>	0.33	81.42	65.17	59.58	59.42
<code>r16rfwftH</code>	0.37	73.25	57.00	56.33	55.92
<code>r17_base</code>	0.20	70.17	77.25	61.67	56.50
<code>r17_houses</code>	0.06	84.50	88.58	69.58	65.00

The pattern is the same everywhere: UU and II are high, UI and IU are low.

Humans do not do this. For humans II (81.88) sits *below* UI (84.13). **Every one of our six models reverses it**, on both face stores — all twelve cells.

What this means. II is the hardest condition for a person: the face is upside down when you learn it and upside down when you are tested. Our models find it nearly as easy as upright. Studying and testing inverted costs `r16_base` 0.17 points on RFW faces. What actually hurts the model is the two orientations *disagreeing*.

So the model is doing something different from a person:

- **Its memory is fine.** It encodes an image and recognises it again through the retrieval noise, upside down or not. Orientation does not make the image harder to store.
- **It does not know the two views are the same face.** Show it upright, test it inverted, and it treats them as different things. A person sees a face they already know, just rotated.

Splitting the 2×2 into main effects plus an interaction makes the gap explicit. The interaction — the extra cost of study and test disagreeing, beyond either main effect — is the term that is wrong:

model	study effect	test effect	mismatch
Human	+9.98	+4.43	+7.73
<code>r15_vgg</code>	+5.13	−1.12	+20.79
<code>r16_base</code>	+5.46	+5.29	+16.12
<code>r16_rfw</code>	+8.21	+8.04	+13.79
<code>r16rfwftH</code>	+8.33	+7.92	+9.00
<code>r17_base</code>	−0.96	−6.12	+14.62

The main effects are roughly human-sized. The mismatch term is two to three times too big.

And it comes from the architecture, not from training. Reading the same battery out of each layer:

readout	study	test	mismatch
layer1	-0.71	-0.29	+ 4.87
layer2	-0.50	-0.58	+4.00
layer3	-0.71	-1.38	+4.46
layer4	+1.75	+4.25	+26.17
h (bottleneck)	+7.17	+4.75	+16.50

(r16rffwH on faces_rfwWM64.) The mismatch cost is **already there at layer1**, where the orientation main effects are exactly zero. Nothing has been learned yet at that depth. It is the log-polar sampling grid: rotate the image 180° and the representation shifts, so matching across a mismatched pair is harder no matter what the image contains. The main effects, by contrast, appear only at layer4 and the bottleneck — those are learned.

One caveat on the stores. faces_rfwWM64 photos are in the wild, so background, lighting, pose and clothing all differ between identities and all survive inversion. The 2AFC can be won without looking at the face — which is why we calibrate on it but measure on faces_cfdWM64.

5. Putting inverted images into the diet

I thought exposing the network to some inverted images might help orientation sensing but that’s obviously a dumb idea you can skip this portion but I still wrote it up. The r18 arms are r16_rfw’s recipe with one thing changed: a fraction of training images are flipped. r16_rfw itself is the 0% control.

I expected the mismatch term to fall toward the human +7.73 while the upright advantage survived — most of the diet is still upright.

What happened is that everything went at once, and fast.

Training accuracy first. The gap between upright and inverted classification:

arm	faces upright / inverted	gap
r16_rfw (0%)	90.47 / 63.31	27.16
r18_inv02 (2%)	86.26 / 85.40	+ 0.86
r18_inv05 (5%)	86.35 / 86.07	+0.28
r18_inv20 (20%)	86.68 / 86.26	+0.42

Two percent of the diet erases a 27-point gap. 5% and 20% add nothing. The effect saturates below the smallest dose we have tried.

The memory task says the same thing, on faces_cfdWM64:

model	UU	II	UI	IU	study	test	mismatch
Human	96.29	81.88	84.13	78.58	+9.98	+4.43	+ 7.73
r16_rfw (0%)	81.42	65.17	59.58	59.42	+8.21	+8.04	+13.79
r18_inv02 (2%)	67.00	72.75	62.33	64.42	-3.92	-1.83	+ 6.50
r18_inv05 (5%)	72.50	78.00	70.58	71.83	-3.37	-2.12	+ 4.04
r18_inv20 (20%)	76.67	78.67	74.92	74.08	-0.58	-1.42	+ 3.17

The four cells collapse together. In the 0% control the two mismatch cells sit about 22 points below UU (59.58 and 59.42 against 81.42). With inverted images in the diet that gap closes to about 4 points at 2% and 2 points at 20%. UI and IU climb — 59.58 → 62.33 → 70.58 → 74.92 — which is exactly what we were trying to achieve. The problem is that UU **falls** at the same time, 81.42 → 76.67, so the four conditions meet in the middle rather than the mismatch cells rising to meet upright. The model ends up equally good at everything, which is not what a person does.

(UU is not monotonic across the arms — 67.00 at 2%, 72.50 at 5%, 76.67 at 20%, all below the 0% control’s 81.42. All four are calibrated to nearly the same noise level, $p = 0.33\text{--}0.36$, so this is not a calibration artefact. Unexplained.)

The idea did not work the way we wanted. The mismatch term did collapse, and cleanly with dose — $+13.79 \rightarrow +6.50 \rightarrow +4.04 \rightarrow +3.17$, straight past the human $+7.73$. But the study and test effects collapsed with it, from $+8.21$ and $+8.04$ down to negative at every dose. II is now *higher* than UU.

The 2% arm is the awkward one. Its mismatch term, **$+6.50$, is the closest to the human $+7.73$ of any model in this document** — better than `r16rfwtH`’s $+9.00$. It gets there with study -3.92 and test -1.83 , so the number is right for the wrong reason: the model is not more human-like, it has simply stopped caring about orientation at all.

A few percent of inverted exposure does not buy partial orientation tolerance. It makes the model close to orientation-blind, which removes every orientation signature at once, including the human-like ones we wanted to keep. There is no dose here where the architectural term drops and the learned advantage survives.

6. Where it stands

	status
Kanwisher, faces	matched — <code>r15_vgg</code> 86.10 / $+11.10$ vs human 87.50 / $+10.70$
Yin, faces, upright advantage	close — <code>r16_rfw</code> and <code>r16rfwtH</code> $+16.25$ vs human $+14.41$
Yin, faces, mismatch cost	$2\text{--}3\times$ too big in every model; comes from the log-polar encoding, not training
Yin, objects	correct — near zero everywhere, but stuck at ceiling
Yin, houses	repaired, but the cost grows with house training, so it is a confounded control
Inverted diet	every dose from 2% up erases the classification gap and the memory effects together; no dose separates them
